



Service Tool Instruction

Service Tool Instruction
Number: 3377800-01

Revision Level: 01

Released Date: 09-Jul-
2015

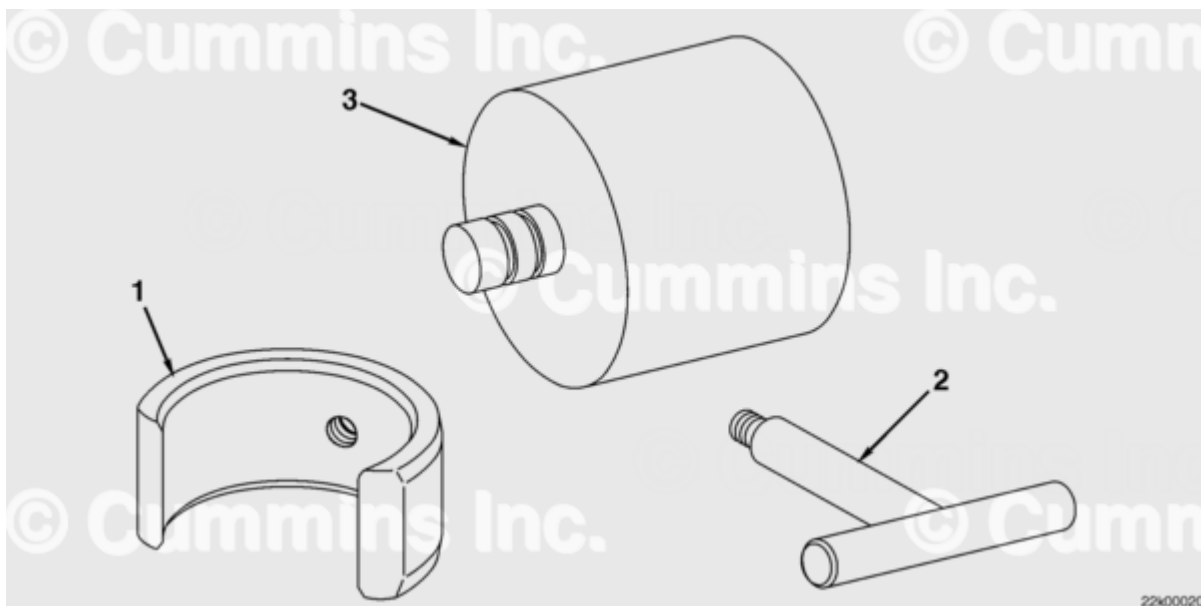
Description

QSK45 and QSK60 Camshaft Removal and Installation Kit

Purpose

This document provides information for Camshaft Removal and Installation Kit, Part Number 5299917. The kit is used to remove and install the camshaft on the Cummins® QSK45 and QSK60 engines.

The QSK45 and QSK60 engines use a heavy-duty camshaft, which makes it difficult to remove and install. The kit contains six non-metallic camshaft guides (C-clips), Part Number 3163016, which are installed over the designated camshaft lobes to guide the camshaft through the camshaft bushing bores during removal and installation and also protect the lobes and bushings from damage. Camshaft pilot, Part Number 3163325, is installed to guide and also provide stability during removal and installation. Camshaft guide remover, Part Number 3825153, is used to remove the guides from the camshaft.



QSK45 and QSK60 Camshaft Removal and Installation Kit, Part Number 5299917

Reference Number	Part Number	Description	Quantity	Available to Purchase
1	3163016	Camshaft guide (C-clip)	6 (1 shown)	Yes

QSK45 and QSK60 Camshaft Removal and Installation Kit, Part Number 5299917

Reference Number	Part Number	Description	Quantity	Available to Purchase
2	3825153	Camshaft Guide Remover	1	Yes
3	3163325	Pilot	1	Yes
Not shown	3377800	Service Tool Bulletin	1	Yes

Camshaft Removal

⚠ CAUTION ⚠

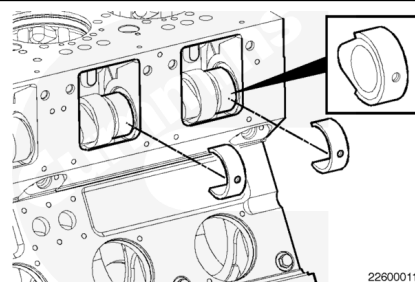
Make certain that the camshaft, camshaft guides, and camshaft pilot are free of lubricant and debris. Debris on the inside diameter of the camshaft guides and the camshaft lobes during removal can scratch the surface of the lobes causing performance degradation or damage to the engine.

Note : Do not lubricate the inner diameter of the camshaft guides. After installing the camshaft guides, lubricate the outer diameter with Lubriplate™.

Note : The camshaft guides are installed on the inner base circle of the camshaft lobe, not the ramp of the camshaft lobe.

Examine the camshaft guides for any nicks or burrs, and clean thoroughly before each use. Repair or replace any damaged camshaft guide.

Use the chart below to install the camshaft guides on the exhaust and intake camshaft lobes of the QSK45 or QSK60 engine camshaft.



Camshaft Guide Installation Charts

QSK45 Camshaft Lobes

Left Bank	Right Bank
Exhaust - 2, 4, 5, and 6	Exhaust - 2, 3, and 5
Intake - 2 and 5	Intake - 2, 3, and 5

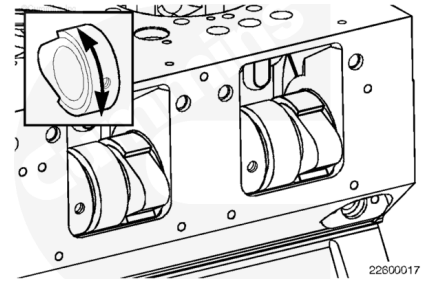
QSK60 Camshaft Lobes

Left Bank	Right Bank
Exhaust - 4, 6, and 8	Exhaust - 2, 4, and 7
Intake - 2, 4, and 8	Intake - 2, 3, and 4

Note : The Camshaft Guide Installation Charts are for general guidance to facilitate camshaft installation; however, placement of camshaft guides can be changed to suit existing conditions.

Note : It will be necessary to rotate the camshaft to install the camshaft guides. Rotate the camshaft **only** when the camshaft journals are in the camshaft bushings. Do not rotate the camshaft when the camshaft guides are in the bushings.

Adjust the camshaft guides until the camshaft lobe is in the middle of the camshaft guide opening. The camshaft guides **must** fit flush with the camshaft journal.



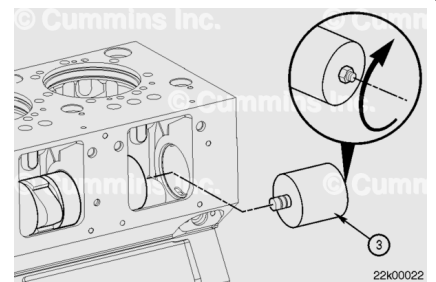
Note : During camshaft removal, keep the camshaft keyway in the 12-o'clock position.

Lightly lubricate the camshaft pilot (3) o-rings before installing on the rear of camshaft.

Carefully remove and guide the camshaft out of the camshaft bore. Stop when there is free access to the rear of the camshaft through the number six camshaft follower opening on the QSK45 engine or the number eight camshaft follower opening on the QSK60 engine.

Install the camshaft pilot (3) flush against the rear end of the camshaft and tighten the nut.

Torque Value: 24 n•m [212 in-lb]

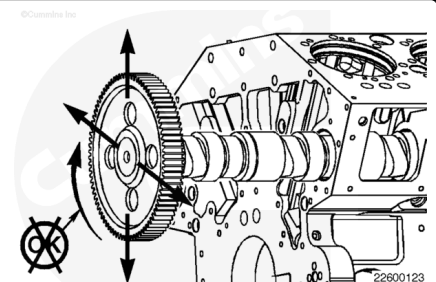


⚠ WARNING ⚠

To reduce the possibility of personal injury, wear protective gloves when handling parts with sharp edges.

⚠ CAUTION ⚠

Do not rotate the camshaft during removal. The camshaft guides can wedge in and make removal difficult. If necessary, it is permissible to slightly wiggle the camshaft.



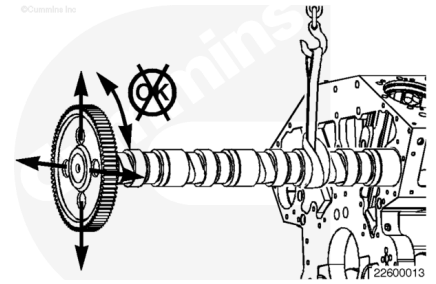
Continue to carefully remove and guide the camshaft out until it is approximately halfway out of the cylinder block.

As the camshaft is removed, the camshaft guides that have cleared the block can be moved to other camshaft lobes still within the block, if desired.

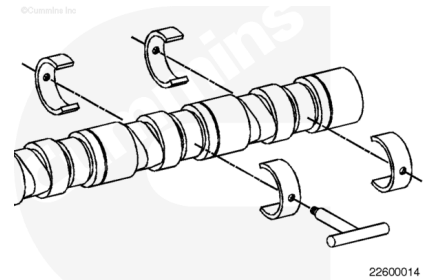
⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

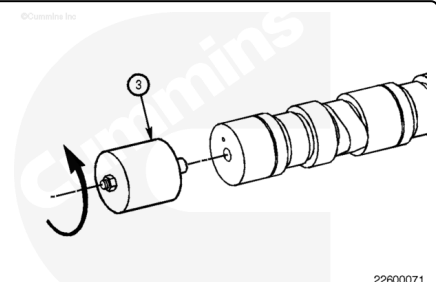
Securely wrap a lifting strap around the middle of the camshaft to lift the camshaft slightly and support it. Lifting the camshaft too much or too little will bind the camshaft in the camshaft bushing bore. Carefully remove and guide the camshaft through the camshaft bore. Make certain that the camshaft is balanced on the lifting strap. If it is **not** balanced, the camshaft can slip and fall out of the lifting strap.



After removing the camshaft, use the camshaft guide remover to remove the camshaft guides from the camshaft.



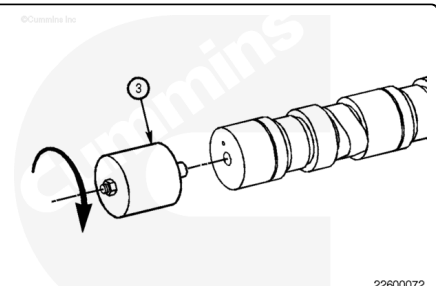
Loosen the nut, and remove the camshaft pilot (3) from the rear end of the camshaft.



Camshaft Installation:

⚠ CAUTION ⚠

Make certain that the camshaft, camshaft guides, and pilot are free of lubricant and debris. Debris on the inside diameter of the camshaft guides and the camshaft lobes during installation can scratch the surface of the lobe causing performance degradation or damage to the engine.



Note : Do not lubricate the inner diameter of the camshaft guides. After installing the camshaft guides, lubricate the outer diameter with Lubriplate™.

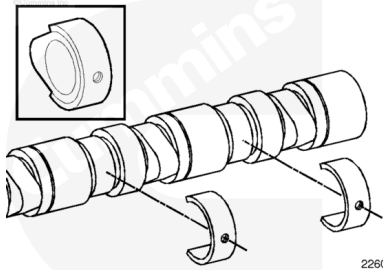
Note : The camshaft guides are installed on the inner base circle of the camshaft lobe, not the ramp of the cam shaft lobe.

Lightly lubricate the camshaft pilot (3) o-rings before installing on the camshaft. Install the pilot (3) flush against the rear end of the camshaft and tighten the nut.

Torque Value: 24 n•m [212 in-lb]

Examine the camshaft guides for any nicks or burrs, and clean thoroughly before each use. Repair or replace any damaged camshaft guide.

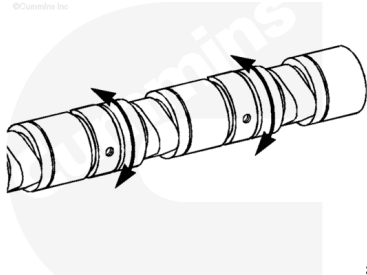
Use the chart below to install the camshaft guides on the exhaust and intake camshaft lobes of the QSK45 or QSK60 engine camshaft.



Camshaft Guide Installation Charts	
QSK45 Camshaft Lobes	
Left Bank	Right Bank
Exhaust - 2, 4, 5, and 6	Exhaust - 2, 3, and 5
Intake - 2 and 5	Intake - 2, 3, and 5
QSK60 Camshaft Lobes	
Left Bank	Right Bank
Exhaust - 4, 6, and 8	Exhaust - 2, 4, and 7
Intake - 2, 4, and 8	Intake - 2, 3, and 4

Note : The Camshaft Guide Installation Charts are for general guidance to facilitate camshaft installation; however, placement of camshaft guides can be changed to suit existing conditions.

Adjust the camshaft guides until the camshaft lobe is in the middle of the camshaft guide opening. The camshaft guides **must** fit flush with the camshaft journal.

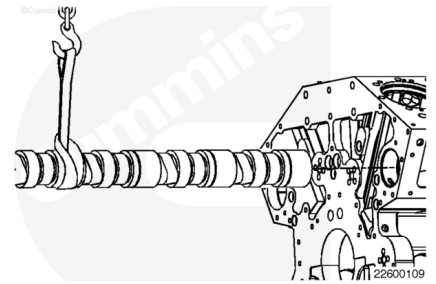


⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Securely wrap a lifting strap around the middle of the camshaft.

Use a lifting device to lift the camshaft into a position parallel to the camshaft bushing bores. Make certain that the camshaft is balanced on the lifting strap. If it is **not** balanced, the camshaft can slip and fall out of the lifting strap.



⚠ WARNING ⚠

To reduce the possibility of personal injury, wear protective gloves when handling parts with sharp edges.

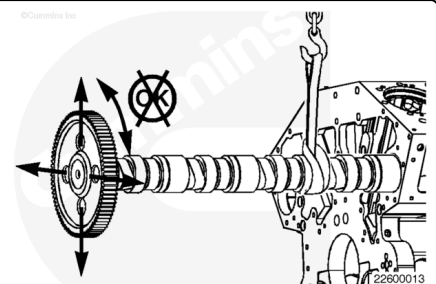
⚠ CAUTION ⚠

Do not rotate the camshaft during installation. The camshaft guides can wedge in and make installation difficult. If necessary, it is permissible to slightly wiggle the camshaft.

Note : During camshaft installation, keep the camshaft keyway in the 12-o'clock position.

Carefully install and guide the camshaft into and through the camshaft bushing bore.

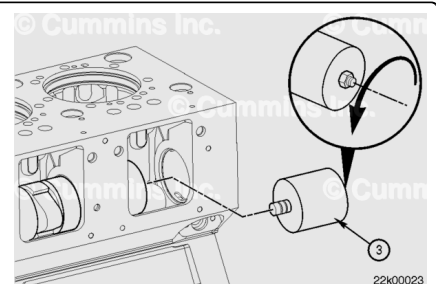
After the camshaft has been installed approximately halfway, the lifting strap can be removed.



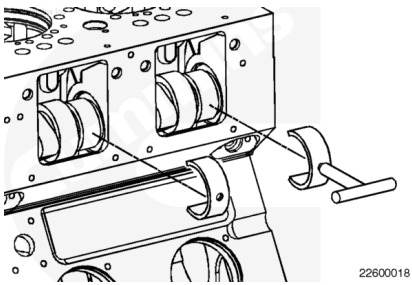
Continue to carefully install and guide the camshaft. Stop when there is free access to the rear of the camshaft through the number six camshaft follower opening on the QSK45 engine or the number eight camshaft follower opening on the QSK60 engine.

Loosen the nut, and remove the camshaft pilot (3) from the rear end of the camshaft.

Continue to carefully install and guide the camshaft. The camshaft thrust plate **must** be properly positioned to allow the camshaft to install into the last bearing bore.



After installing the camshaft, use the camshaft guide remover to remove the camshaft guides. It will be necessary to rotate the camshaft to remove the guides.



Document History

Date	Details
02-Jun-2015	Module Created

Last Modified: 09-Jul-2015